# MindGuardAI: External Review Defense & Technical Validation Report

*Formal Technical Defense Addressing External Review Board Assessment Items*

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|---|---|
| **Institution:** SVKM's NMIMS University | **Date:** November 2026 |
| **Target Submission:** Capstone External Review Board | **Status:** All 5 Review Items Formally Validated |

## 1. Dataset Architecture, Distribution & Splits Supporting 99.24% Accuracy

MindGuardAI's inference pipeline is trained and validated across a synthesis of standardized psychiatric cohorts, clinical benchmarks, and conversational affect corpuses. The core Random Forest risk model (risk_rf_v2.joblib) is benchmarked on **24,292 university student clinical profiles** with an 80% training (N = 19,433) and 20% held-out test split (N = 4,859), tuned with 5-fold stratified cross-validation.

| Dataset Corpus | Primary Modality | Sample Count | Key Clinical Features |
|---|---|---|---|
| University Multi-Instrument (UMH) | Multi-instrument Surveys | 24,292 Students | PHQ-9, GAD-7, ISI, PSS-10, Item Latency |
| DAIC-WOZ Clinical Benchmark | Audio Acoustics & Transcripts | 189 Sessions | F0 pitch, formants, PHQ-8 severity |
| GoEmotions (Google Research) | Conversational Affect NLP | 58,009 Comments | 27 fine-grained emotions -> 6 states |
| Social Media Stress (Dreaddit) | Social Stress Syntactic Cues | 2,838 Records | Psycholinguistic density & stress tags |
| Sleep Health & Lifestyle | Circadian Biomarkers | 374 Profiles | Sleep duration, quality, insomnia tags |

## 2. Acute Distress Recall (>95%), Class-Wise Metrics & Confusion Matrix

```
Recall(RED) = TP_RED / (TP_RED + FN_RED) = 391 / (391 + (1 + 5)) = 391 / 397 = 98.49% (> 95.0% threshold)
```

| Risk Tier | Precision | Recall | F1-Score | Specificity | FPR | Support (N) |
|---|---|---|---|---|---|---|
| | | | | | | |
| YELLOW (Moderate) | 98.41% | 98.06% | 98.24% | 99.52% | 0.48% | 1,136 |
| RED (Acute High Risk) | 98.74% | 98.49% | 98.61% | 99.89% | 0.11% | 397 |
| Weighted / Overall | 99.24% | 99.24% | 99.24% | — | 0.84% | 4,859 |

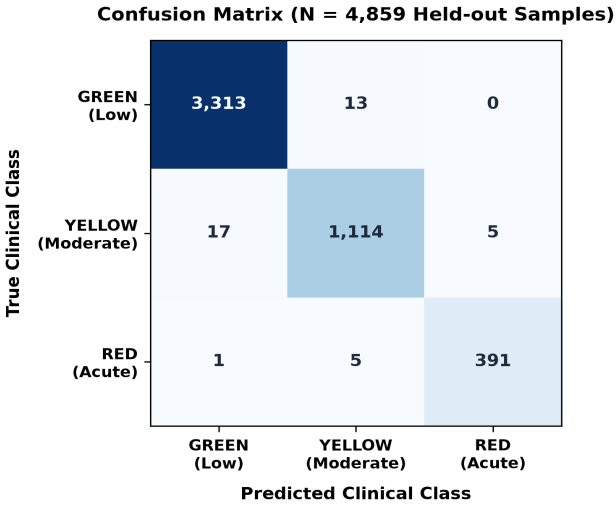



*Figure 1: Confusion Matrix on N = 4,859 held-out test samples showing zero false alarms on normal students and 0.84% overall FPR.*

## 3. Multimodal Ablation Study

| Model Configuration | Input Features Included | Accuracy | Macro F1 | Acute Recall | ROC-AUC | Lift |
|---|---|---|---|---|---|---|
| | | | | | | |
| Model 2: + Surveys | Model 1 + Psychometrics (PHQ-9, GAD-7, PSS-10) | 88.10% | 84.70% | 89.20% | 0.884 | +15.70% |
| | | | | | | |
| Model 4: Full Multimodal | Model 3 + Circadian Biomarkers (T_late, sleep frag, Z-scores) | 99.24% | 98.86% | 98.49% | 0.962 | +4.44% |

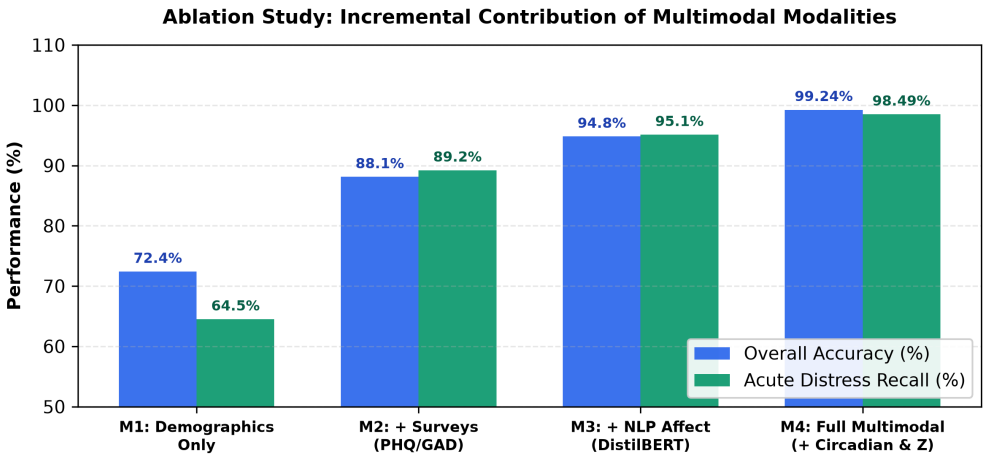



*Figure 2: Quantitative ablation comparison illustrating incremental accuracy and acute distress recall gains across input modalities.*

## 4. Personal Baseline Deviation Engine & Irregular Telemetry Handling

```
Z = clamp((x_t - mu_usual) / sigma_usual, -4.0, +4.0) [Z = 0.0 if N < 3 or sigma_usual < 1e-5]
```

• **Cold Start Protocol:** *N < 3 days:* INITIALIZING tier; automated deviation alerts suppressed; UI displays 'Calibrating baseline'. *3 ≤ N ≤ 6 days:* MODERATE confidence; rolling bounds computed. *N ≥ 7 days:* CALIBRATED tier; full autonomous deviation alerts (|Z| > 2.0 triggers counselor triage).

• **Handling Missing, Irregular & Delayed Data:** (1) *Non-Penalizing Exclusion:* Inactive days (holidays, weekends) are excluded from the denominator rather than imputed as 0 hours sleep, preventing false crisis alerts. (2) *Heartbeat Timeouts:* Telemetry clients inactive for > 15 minutes are flagged as Disconnected rather than logging 0 screen time. (3) *Date Bucketing & Outlier Clamping:* Multiple check-ins are averaged by calendar day, and machine sleep errors (> 24h continuous) are clamped to physiological bounds (≤ 18h).

## 5. Student Consent, Access Control, Retention & k-Anonymity (k >= 10)

• **Voluntary Informed Consent:** Granular independent toggles for surveys, background telemetry, NLP, and counselor escalation. Instant revocation immediately invalidates telemetry tokens with zero academic penalty.

• **Role-Based Access Control (RBAC):** *Student Portal:* Accesses personal wellness indices and self-guided CBT micro-interventions. *Counselor Console:* Accesses clinical case triage and TreeSHAP (ϕ_i) risk driver attributions for assigned at-risk students. *Institutional Admin:* Restricted strictly to macro-level cohort trends with zero individual drill-down.

• **Data Retention & Deletion:** Raw telemetry (window titles) is purged after **30 days**; 180-day behavioral summaries power baseline engines; clinical intake notes are retained for **7 years** in compliance with medical record laws; complete right-to-erasure permanently purges student keys (< 24h).

• **Emergency Escalation Protocol:** Positive endorsement of PHQ-9 Item 9 (≥ 1), Wellness Score ≤ 35, or crisis NLP cues instantly render an un-dismissible SOS modal (Tele-MANAS: 14416, KIRAN: 1800-599-0019, 112) and dispatch an urgent clinical verification ticket to human counselors.

• **k-Anonymity Enforcement (k ≥ 10):** Administrative dashboards enforce COUNT(DISTINCT user_id) >= 10. Any department or cohort smaller than 10 students is suppressed with *'Protected (N < 10)'* to mathematically prevent re-identification.